

Homework Assignment 2 (Rewrite)

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1. (1.A.8) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Prove that

$$\int_a^b f = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{j=1}^n f \left(a + \frac{j(b-a)}{n} \right). \quad (1)$$

Proof. Define a partition $P_n = \left\{ a, a + \frac{b-a}{n}, \dots, a + \frac{(n-1)(b-a)}{n}, b \right\}$ for each $n \in \mathbb{Z}^+$. Let $\varepsilon > 0$ and choose P such that $L(f, P, [a, b]) > \int_a^b f - \frac{\varepsilon}{2}$.

Then

$$\int_a^b f - \frac{\varepsilon}{2} < L(f, P, [a, b]) \leq L(f, P \cup P_n, [a, b]). \quad (2)$$

Let α be the number of points in P . Since P_n and $P \cup P_n$ agree on all subintervals of P_n not containing a point of P , their difference is entirely determined by the behavior on subintervals containing a point of P , of which there are at most α , and each of which has length $\frac{1}{n}$. Since f is Riemann integrable, it is bounded, so let $U = \sup f$ and $L = \inf f$. Then any infimum along a subinterval is greater than or equal to L , and any supremum along a subinterval is less than or equal to U . Thus we obtain an upper bound on the difference of the lower sums:

$$L(f, P \cup P_m, [a, b]) - L(f, P_m, [a, b]) \leq \frac{\alpha}{n}(U - L).$$

Since α is fixed, we can choose an m large enough such that

$$\begin{aligned} L(f, P \cup P_m, [a, b]) - L(f, P_m, [a, b]) &< \frac{\varepsilon}{2} \\ \implies -\frac{\varepsilon}{2} &< L(f, P_m, [a, b]) - L(f, P \cup P_m, [a, b]). \end{aligned}$$

Adding this to (2), we get

$$\int_a^b f - \varepsilon < L(f, P_m, [a, b]).$$

An entirely analogous argument gives us an m such that

$$\int_a^b f + \varepsilon > U(f, P_m, [a, b]).$$

Finally, we observe that since

$$\inf_{\left[a + \frac{(j-1)(b-a)}{n}, a + \frac{j(b-a)}{n} \right]} f \leq f \left(a + \frac{j(b-a)}{n} \right) \leq \sup_{\left[a + \frac{(j-1)(b-a)}{n}, a + \frac{j(b-a)}{n} \right]} f,$$

$$L(f, P_n, [a, b]) \leq \frac{b-a}{n} \sum_{j=1}^n f \left(a + \frac{j(b-a)}{n} \right) \leq U(f, P_n, [a, b]),$$

and thus since we showed that the lower and upper sums over P_n create sequences which limit to $\int_a^b f$, the middle sum above is squeezed so that (1) is satisfied. $\square$

2. (1.A.12) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Prove that $|f|$ is Riemann integrable and that

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Proof. We begin by considering the relationship between $\sup_{[x,y]} |f| - \inf_{[x,y]} |f|$ and $\sup_{[x,y]} f - \inf_{[x,y]} f$ for any interval $[x, y]$. In particular, we want to prove that $\sup_{[x,y]} |f| - \inf_{[x,y]} |f| \leq \sup_{[x,y]} f - \inf_{[x,y]} f$. If $\inf_{[x,y]} f \geq 0$, then $|f| = f$, so the expression is an equality. If $\sup_{[x,y]} f \leq 0$, then $|f| = -f$, so $\sup_{[x,y]} |f| = -\inf_{[x,y]} f$ and $\inf_{[x,y]} |f| = -\sup_{[x,y]} f$, so

$$\sup_{[x,y]} |f| - \inf_{[x,y]} |f| = -\inf_{[x,y]} f + \sup_{[x,y]} f$$

and we again have an equality. The final case occurs when $\sup_{[x,y]} f \geq 0$ but $\inf_{[x,y]} f < 0$. In this case, either $\sup_{[x,y]} |f| = -\inf_{[x,y]} f$ or $\sup_{[x,y]} |f| = \sup_{[x,y]} f$. If the former is true, then we can note that $\sup_{[x,y]} |f| = \sup_{[x,y]} |-f| = \sup_{[x,y]} (-f)$ and

$$\sup_{[x,y]} |f| - \inf_{[x,y]} |f| = \sup_{[x,y]} |-f| - \inf_{[x,y]} |-f|,$$

so we can consider the latter case for $-f$ and get an equivalent result. Having reduced it to the case where $\sup_{[x,y]} |f| = \sup_{[x,y]} f$, we observe that $\inf_{[x,y]} f \leq \inf_{[x,y]} |f|$, so

$$\begin{aligned} \sup_{[x,y]} |f| - \inf_{[x,y]} |f| &\leq \sup_{[x,y]} |f| - \inf_{[x,y]} f \\ &= \sup_{[x,y]} f - \inf_{[x,y]} f. \end{aligned}$$

Having proved that in every case $\sup_{[x,y]} |f| - \inf_{[x,y]} |f| \leq \sup_{[x,y]} f - \inf_{[x,y]} f$, we let $\varepsilon > 0$ and choose a partition $P = \{a = x_0, x_1, \dots, x_n = b\}$ such that

$$U(f, P, [a, b]) - L(f, P, [a, b]) < \varepsilon.$$

Then

$$\begin{aligned} U(|f|, P, [a, b]) - L(|f|, P, [a, b]) &= \sum_{i=1}^n (x_i - x_{i-1}) \sup_{[x_{i-1}, x_i]} |f| - \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} |f| \\ &= \sum_{i=1}^n (x_i - x_{i-1}) \left(\sup_{[x_{i-1}, x_i]} |f| - \inf_{[x_{i-1}, x_i]} |f| \right) \\ &\leq \sum_{i=1}^n (x_i - x_{i-1}) \left(\sup_{[x_{i-1}, x_i]} f - \inf_{[x_{i-1}, x_i]} f \right) \\ &= U(f, P, [a, b]) - L(f, P, [a, b]) \\ &< \varepsilon. \end{aligned}$$

Thus $|f|$ is Riemann integrable since ε is arbitrary.

For the inequality, we want to show the equivalent statement that

$$-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|.$$

The second inequality comes from the fact that for any interval $[\alpha, \beta]$ and any partition P of $[a, b]$,

$$\begin{aligned} f(x) &\leq |f(x)| \quad (\text{for all } x) \\ \implies \inf_{[\alpha, \beta]} f &\leq \inf_{[\alpha, \beta]} |f| \\ \implies L(f, P, [a, b]) &\leq L(|f|, P, [a, b]) \\ \implies L(f, [a, b]) &\leq L(|f|, [a, b]) \\ \implies \int_a^b f &\leq \int_a^b |f|. \end{aligned}$$

The first inequality comes from the fact that $-\inf_{[x_{i-1}, x_i]} |f| \leq \inf_{[x_{i-1}, x_i]} f$ since $-|f(x)| \leq f(x)$ for all x , so $L(-|f|, P, [a, b]) \leq L(f, P, [a, b])$. $\square$

3. (1.B.1) Define $f : [0, 1] \rightarrow \mathbb{R}$ as follows:

$$f(a) = \begin{cases} 0 & \text{if } a \text{ is irrational,} \\ \frac{1}{n} & \text{if } a \text{ is rational and } n \text{ is the smallest positive integer} \\ & \text{such that } a = \frac{m}{n} \text{ for some integer } m. \end{cases}$$

Show that f is Riemann integrable and compute $\int_0^1 f$.

Proof. For any partition, the lower Riemann sum is 0 since there is an irrational number contained within every subinterval. Thus, we will show that we can find a partition P_n for which $U(f, P_n, [0, 1]) < \frac{1}{n}$, allowing us to conclude that $U(f, [0, 1]) = 0$.

We begin with the case $n = 2$. Let $\varepsilon > 0$ be small enough that $[\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon]$ does not contain $\frac{m}{3}$ where $0 \leq m \leq 3$ and $\frac{1}{3} + \frac{1}{3}\varepsilon < \frac{1}{2}$. Consider the partition

$$P_2 = \left\{ 0, \frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon, 1 \right\}.$$

Then

$$\begin{aligned} U(f, P_2, [0, 1]) &= \left(\frac{1}{2} - \varepsilon \right) \cdot \frac{1}{3} + 2\varepsilon \cdot \frac{1}{2} + \left(\frac{1}{2} - \varepsilon \right) \cdot \frac{1}{3} \\ &= \left(\frac{1}{2} - \varepsilon \right) \cdot \frac{2}{3} + \varepsilon \\ &= \frac{1}{3} - \frac{2}{3}\varepsilon + \varepsilon \\ &= \frac{1}{3} + \frac{1}{3}\varepsilon \\ &\leq \frac{1}{2}. \end{aligned}$$

We can follow a similar procedure for a general P_n , choosing partition points close enough to $\frac{m}{\ell}$ for all $\ell \leq n$ and $0 \leq m \leq \ell$ so that the upper sum is determined up to some fraction of ε by the larger subintervals, whose maximum value is at most $\frac{1}{n+1}$. $\square$

4. (1.B.4) Give an example of bounded functions $f, g : [0, 1] \rightarrow \mathbb{R}$ such that

$$L(f, [0, 1]) + L(g, [0, 1]) < L(f + g, [0, 1])$$

and

$$U(f + g, [0, 1]) < U(f, [0, 1]) + U(g, [0, 1]).$$

Proof. Define

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

and

$$g(x) = \begin{cases} -1 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then $(f + g)(x) = 0$ for all $x \in [0, 1]$, so

$$L(f, [0, 1]) + L(g, [0, 1]) = -1 - 1 = -2 < 0 = L(f + g, [0, 1])$$

and

$$U(f, [0, 1]) + U(g, [0, 1]) = 1 + 1 = 2 > 0 = U(f + g, [0, 1]).$$

$\square$

5. (2.A.1) Prove that if A and B are subsets of $\mathbb{R}$ and $|B| = 0$, then $|A \cup B| = |A|$.

Proof. By countable subadditivity,

$$|A \cup B| \leq |A| + |B| = |A| + 0 = |A|.$$

By monotonicity, since $A \subset A \cup B$,

$$|A| \leq |A \cup B|.$$

Therefore, $|A \cup B| = |A|$. $\square$

6. (2.A.8) Prove that if $A \subset \mathbb{R}$ and $t > 0$, then $|A| = |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))|$.

Proof. By countable subadditivity,

$$|A| \leq |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))|.$$

For the other direction, note that

$$|A| = |A \setminus \{-t, t\}|$$

by 2.A.1 since $|\{-t, t\}| = 0$. Then

$$\begin{aligned} |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))| &= |A \cap (-t, t)| + |(A \cap (-\infty, -t]) \cup (A \cap [t, \infty))| \\ &= |A \cap (-t, t)| + |(A \cap (-\infty, -t)) \cup (A \cap (t, \infty))| \\ &\leq |A \cap (-t, t)| + |A \cap (-\infty, -t)| + |A \cap (t, \infty)|. \end{aligned}$$

Let $\varepsilon > 0$ and let $\{I_n\}_{n \in \mathbb{Z}}$ be a collection of open intervals such that $A \subset \bigcup \{I_n\}$ and

$$|A| + \varepsilon \geq \sum_{n=1}^{\infty} \ell(I_n).$$

Let $J_n = I_n \cap (-\infty, t)$, $K_n = I_n \cap (-t, t)$, and $L_n = I_n \cap (t, \infty)$, each of which is either empty or an open interval. For each n , J_n , K_n , and L_n are pairwise disjoint, and each is either an open interval or the empty set (which will not cause problems for our purposes). Since $J_n \cup K_n \cup L_n \subset I_n$,

$$\ell(I_n) \geq \ell(J_n) + \ell(K_n) + \ell(L_n),$$

and since

$$\begin{aligned} A \cap (-\infty, -t) &\subset \bigcup J_n, \\ A \cap (-t, t) &\subset \bigcup K_n, \text{ and} \\ A \cap (t, \infty) &\subset \bigcup L_n, \end{aligned}$$

we have

$$\begin{aligned} |A| + \varepsilon &\geq \sum_{n=1}^{\infty} \ell(I_n) \\ &\geq \sum_{n=1}^{\infty} \ell(J_n) + \sum_{n=1}^{\infty} \ell(K_n) + \sum_{n=1}^{\infty} \ell(L_n) \\ &= \sum_{n=1}^{\infty} |J_n| + \sum_{n=1}^{\infty} |K_n| + \sum_{n=1}^{\infty} |L_n| \\ &\geq \left| \bigcup_{n=1}^{\infty} J_n \right| + \left| \bigcup_{n=1}^{\infty} K_n \right| + \left| \bigcup_{n=1}^{\infty} L_n \right| \\ &\geq |A \cap (-\infty, -t)| + |A \cap (-t, t)| + |A \cap (t, \infty)| \\ &= |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))|. \end{aligned}$$

Since this is true for all $\varepsilon > 0$, we have

$$|A| \geq |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))|.$$

□

What topic do you understand well?

I'd say my understanding of outer measure is the strongest.

What topic or problem has been most difficult for you?

Riemann integration. The process is just so tedious, and I often find myself desperately wanting to find a more elegant way to state something, when I probably just need to get my hands dirty and realize that at least at this level in analysis, things are often unavoidably ugly. I'm looking forward to getting to the more elegant material :). Dealing with supremums and infimums very heavily has taken some getting used to as well, but I think I'm most of the way there. On this homework in particular, I had a lot of trouble with all but the fourth and fifth problems. The thing that probably sticks out most to me was thinking about absolute values. I spent hours trying to prove $\sup |f| - \inf |f| < \sup f - \inf f$.

Which one of the suggested problems that has not appeared on homework would you most like to review?

I haven't had a chance to look at them very closely yet, as I've been racing to finish the homeworks.